544. Output Contest Matches

Medium 268 90 Add to List Share

During the NBA playoffs, we always arrange the rather strong team to play with the rather weak team, like make the rank 1 team play with the rank n_{th} team, which is a good strategy to make the contest more interesting. Now, you're given **n** teams, you need to output their **final** contest matches in the form of a string.

The **n** teams are given in the form of positive integers from 1 to n, which represents their initial rank. (Rank 1 is the strongest team and Rank n is the weakest team.) We'll use parentheses('(', ')') and commas(',') to represent the contest team pairing - parentheses('(' , ')') for pairing and commas(',') for partition. During the pairing process in each round, you always need to follow the strategy of making the rather strong one pair with the rather weak one.

Example 1:

```
Input: 2
Output: (1,2)
Explanation:
Initially, we have the team 1 and the team 2, placed like: 1,2.
Then we pair the team (1,2) together with '(', ')' and ',', which is the final answer.
```

Example 2:

```
Input: 4
Output: ((1,4),(2,3))
Explanation:
In the first round, we pair the team 1 and 4, the team 2 and 3 together, as we need to make the strong team and
weak team together.
And we got (1,4),(2,3).
In the second round, the winners of (1,4) and (2,3) need to play again to generate the final winner, so you need to
add the paratheses outside them.
And we got the final answer ((1,4),(2,3)).
```

Example 3:

```
Input: 8
Output: (((1,8),(4,5)),((2,7),(3,6)))
Explanation:
First round: (1,8),(2,7),(3,6),(4,5)
Second round: ((1,8),(4,5)),((2,7),(3,6))
Third round: (((1,8),(4,5)),((2,7),(3,6)))
Since the third round will generate the final winner, you need to output the answer (((1,8),(4,5)),((2,7),(3,6))).
```

Note:

1. The **n** is in range [2, 2^{12}].
2. We ensure that the input **n** can be converted into the form 2^k, where k is a positive integer.

Accepted 20,368 Submissions 27,135

Seen this question in a real interview before? Yes No

Contributor

Companies

Related Topics

Java Autocomplete

Need more space? Try using full screen mode!

Got it!

1

Your previous code was restored from your local storage.

Console Contribute

Run Code Submit